MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 4J38

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Fixed nominal frequency 3,575 MHz, pulsed, external magnet, rated peak power output = 700 kw.

ABSOLUTE RATINGS:

Parameter:	Ef	tk	epv	ib	pi	Pi	Du	tpc	T(anode)
Unit:	V	sec	kv	a	kw	W	---	μs	°C
Maximum:	17.6	---	30	70	2,000	1,200	0.001	2.5	100
Minimum:	---	120	---	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1

TEST CONDITIONS:

Parameter:	Ef	tk	tpc	Du	trv	Ib	epy	VSWR	H
Unit:	V	sec	μs	---	μs	mAdc	kv	---	Gauss
Tolerance:	---	Max	±0.1	---	Max	---	±2	Max	---
	10.0 (Note 1)	120	1.0	0.0005	0.2	35	28	1.15:1	2,500 (Note 3)

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
4027	Temperate coefficient			ΔF	---	0.07	MHz / °C
1047	Low-temperature operation		tk = 180 seconds (max)	---	---	---	---
1031	High-frequency vibration		No voltages applied	---	---	---	---
	Quality conformance inspection, part 1	2					
4289	Heater current, nonoperating	-	Ef = 16.0 V; tk = 120	If	2.8	3.4	A
4003	Pressurizing	-	40 to 45 psia	---	---	---	---
---	Fixed frequency	-	Lower limit Upper limit	F F	--- 3,550	3,600 ---	MHz MHz
4250	Power output	-	t = 300	Po	350	---	W
4308	RF bandwidth	-		BW	---	2.5/tpc	MHz
4315	Stability	-	Ib = 37.5 mAdc	MP	---	1	%
	Quality conformance inspection, part 2						
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---
4310	Frequency pulling figure	-	Ib = 20 mAdc	ΔF	---	10	MHz
	Quality conformance inspection, part 3						
---	Life test	-	Group D	t	500	---	hrs
---	Life-test end points:						
4250	Power levels	-		Po	320	---	W
4308	RF bandwidth	-		BW	2.5/tpc	---	MHz

NOTES:

1. The TUT heater shall be operated during high-voltage operation according to the following schedule.

Pi (watts)	Ef (volts)
1,000 to 1,200	8
800 to 1,000	10.5
600 to 800	13
400 to 600	15
Less than 400	16

The above schedule is valid only for repetition rates of 300 pps, or greater.

2. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

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NOTES: -Continued

3. The magnetic field shall be measured in accordance with method 1367.

4. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3029)

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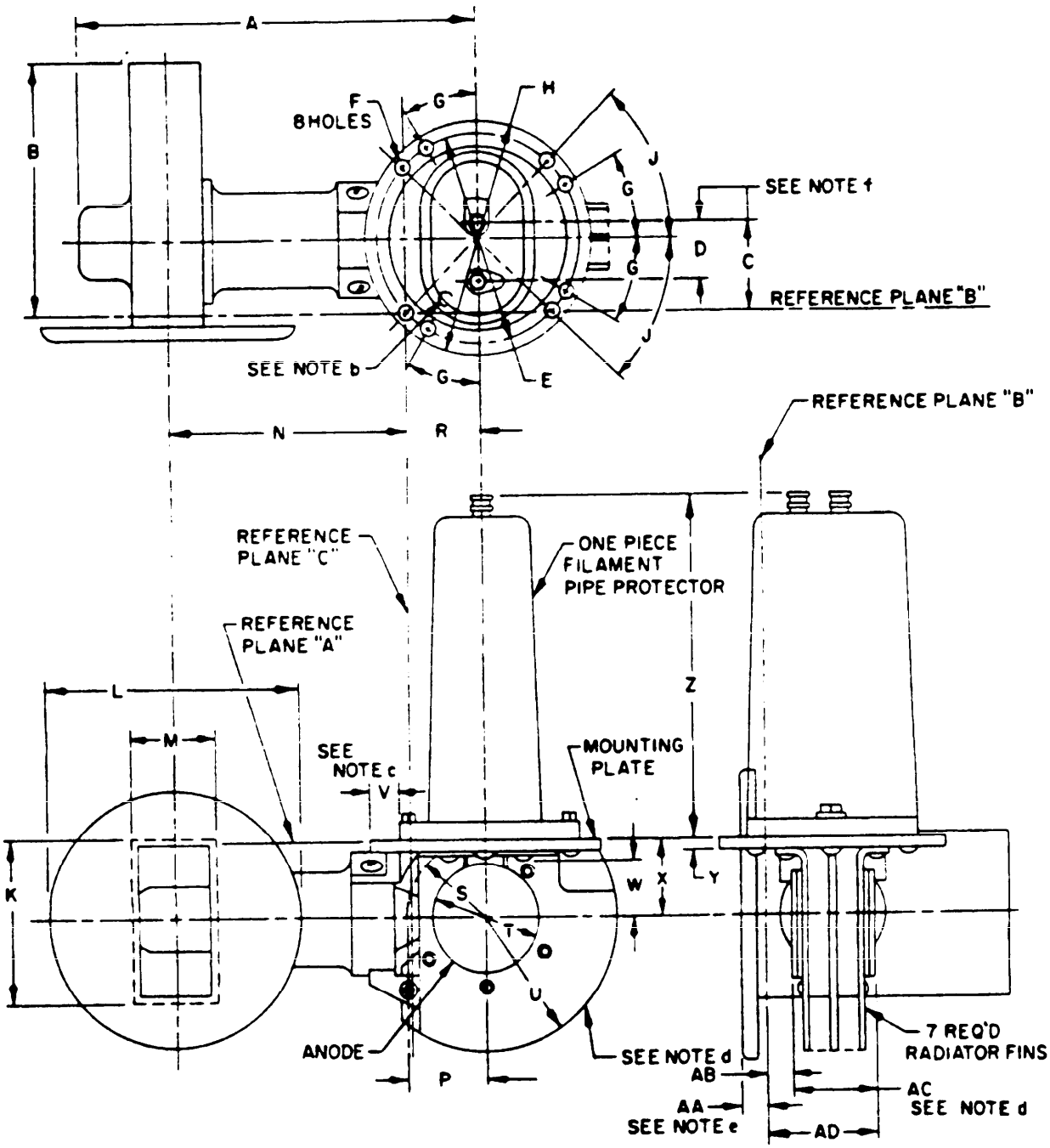


FIGURE 1. Outline drawing of electron tube type 4J38.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			
C	1.515 (38.48) Nom		f
D	.782 (19.86)	.812 (20.62)	f
K	3.000 (76.20) Nom		
L	5.27 (133.86)	5.36 (136.14)	
M	1.500 (38.10) Nom		
N	4.688 (119.08)	4.788 (121.62)	
R	1.44 (36.58) Nom		
T	2.06 (52.32) Nom		
X	1.44 (36.58) Nom		
Y	.19 (4.83) Nom		
AA	.333 (8.46)	.433 (11.00)	e
AB	.677 (17.20)		
AC		1.490 (37.85)	d
AD		2.197 (55.80)	
Quality conformance inspection, part 2			
A		8.23 (209.04)	
B		5.227 (132.77)	
E	4.056 (103.07)	4.070 (103.38)	
F	.205 (5.21)	.215 (5.46)	
G	29°48'	30°12'	
H	4.55 (115.06)	4.59 (116.59)	
J	44°48'	45°12'	
P	1.50 (38.10)		
S	1.50 (38.10)		
U		2.66 (67.56)	
V	.50 (12.70)		c
W	1.06 (26.92)		
Z	6.22 (157.99)	6.41 (162.81)	

NOTES:
a. Paint with a heat resistant noncorrosive paint. The following shall be free from paint: Top surface and parts above mounting plate, inside area, flange face and edge of waveguide.
b. Common cathode connection marked with letter 'C'.
c. This annular area shall be flat within .015(.38 mm). When resting on a flat surface a thickness gage .125(3.18 mm) inch wide shall not enter more than .250(6.35 mm) inch at any point.
d. Maximum width specified applies to area defined by broken line and circumference of radiator.
e. Tolerance includes angular and lateral deviations of the surface.
f. The jackholes shall be within a radius of .100(2.54 mm) of true location.
g. All solder joints on mounting plate shall be soldered to provide hermetic seal.

FIGURE 1. Outline drawing of electron tube type 4J38 - Continued.

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 86 1426

REPLACES EDITION OF 1 OCT 84 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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